

Private Cloud

- It can be designed and operated by one organization
- It can be hosted internally or externally by internal teams or third party
- A private cloud can be build using a software stack like OpenStack To read about [[Hypervisor]]

Public Cloud

- It is open to public and anybody can use it
- Amazon Web Services, Google Cloud Platform, and Microsoft Azure are examples of public clouds. ### Hybrid Cloud Public and Private cloud bound together to form hybrid cloud
- Storage of sensitive information on the private cloud, while offering public services based on that information from a public cloud.
- Meeting temporary resources needs during peak or times of high demand by scaling to the cloud, known as bursting, when such temporary needs of additional computing resources cannot be met by the private cloud. ### Community Cloud Formed by multiple organisations sharing the common cloud infrastructure

Distributed Cloud

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A distributed cloud is a cloud architecture where computing resources are spread across multiple physical locations, such as on-premises, other cloud providers, or edge locations, but are centrally managed from a single control plane. - This model is used to improve performance by reducing latency, meet data residency and regulatory requirements, and support edge computing applications

Multicloud

Multicloud is ==the strategic use of computing services from two or more cloud providers to run an organization's applications and workloads==. This approach allows a company to select the best services from different providers, such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud, based on specific business needs.

By contrast, a **hybrid cloud** combines a public cloud with a private cloud or on-premises infrastructure. A multi-cloud deployment is not necessarily a hybrid one, although a hybrid cloud that uses multiple public clouds would be considered a hybrid multi-cloud.

- **Vendor flexibility and avoiding lock-in:** Organizations can avoid being dependent on a single cloud vendor and can more easily switch providers if needed. This increases negotiation power and flexibility in service choices.
- **Best-of-breed services:** Companies can select the most innovative and suitable services for each specific task. For example, a business might use Google Cloud for machine learning but AWS for its robust storage services.

- **Improved resilience and redundancy:** By distributing workloads across multiple cloud platforms, an organization can minimize the risk of downtime from an outage. If one provider fails, the workload can be failed over to another.
- **Cost optimization:** A multi-cloud strategy allows companies to leverage competitive pricing models and take advantage of cost savings by choosing the most affordable provider for each workload.
- **Geographic reach and compliance:** Companies can use providers with different data center locations to reduce latency for global users and meet regional data sovereignty and compliance requirements. ### Poly Cloud “Poly cloud” refers to using multiple cloud providers where each provider is chosen for its specialized services, unlike a general “multi-cloud” strategy which can be more vendor-agnostic. For example, an organization might use Google Cloud for data analytics and Microsoft Azure for Office 365 integration.

This approach allows businesses to leverage the best capabilities of each provider for specific tasks, leading to performance gains and reduced risk from a single vendor.

Specialization:

A poly cloud model actively chooses a cloud provider based on its specialized strengths for a particular service, such as a provider with a superior machine learning platform or a specific database service.

- **Best-of-breed approach:** It aims to use the best tool for the job by mixing and matching services from different clouds, rather than trying to make applications portable across all providers.
- **Differentiation:** It encourages cloud providers to differentiate themselves through higher-order capabilities, which poly cloud utilizes.
- **Hybrid cloud integration:** The concept can also apply to hybrid environments where an organization uses a mix of public cloud and private cloud services. — Read next chapter [[Virtualization]]